

Fake News Detection - Project Documentation

Overview

This project focuses on building a machine learning-based system to classify news articles as either **real** or **fake**. By processing textual data from news articles, the system learns patterns and indicators commonly found in misleading versus factual content. Multiple classification models are applied and evaluated to determine the most effective approach.

Dataset

Two separate datasets were used:

- **real.csv**: Contains genuine news articles.
- **fake.csv**: Contains fabricated or misleading news articles.

Each dataset includes various fields such as title, text, subject, and publication date.

Data Preprocessing

1. Loading and Labeling Data

The data was initially loaded and labeled by adding a new `class` column:

- Real news was labeled with 1.
- Fake news was labeled with 0.

2. Manual Testing Set

The last 10 rows from both datasets were extracted and saved for manual testing purposes. These rows were then excluded from the training and evaluation process.

3. Merging Datasets

After labeling and removing the manual testing rows, the real and fake news datasets were merged into a single combined dataset for processing.

4. Column Cleanup

Non-essential columns such as `title`, `subject`, and `date` were removed to retain only the main textual content required for analysis.

5. Text Cleaning

Text was preprocessed using regular expressions to:

- Convert all text to lowercase
- Remove punctuation and special characters
- Standardize formatting

This step ensured that the input data was clean and consistent for model training.

Train-Test Split

The cleaned dataset was divided into training and testing sets using a standard 75/25 split. This allows the model to learn from the training data and be evaluated on unseen data.

Text Vectorization

To convert the textual data into a format suitable for machine learning, TF-IDF (Term Frequency-Inverse Document Frequency) vectorization was applied. This technique quantifies the importance of words relative to the document and corpus, helping models understand the significance of different terms.

Model Training & Evaluation

Four different machine learning classifiers were trained on the dataset. Each model was evaluated using accuracy and a classification report (precision, recall, F1-score):

1. Logistic Regression

A linear model commonly used for binary classification. It performed well in detecting patterns in text data.

2. Decision Tree Classifier

A simple yet interpretable model that splits data based on feature values. Its performance was evaluated but is generally prone to overfitting.

3. Gradient Boosting Classifier

An ensemble model that builds trees sequentially and focuses on reducing errors. It typically provides strong accuracy and robustness.

4. Random Forest Classifier

An ensemble of decision trees trained on random subsets of the data. It offers good generalization and accuracy for classification tasks.

User-Defined Functions

Two helper functions were created:

- **Label Interpreter:** Converts numerical predictions into human-readable labels:
 - 0 → "Fake News"
 - 1 → "Not a Fake News"
- **Manual Testing:** Designed to test and interpret the reserved manual samples using the trained model.

Conclusion

The Fake News Detection system effectively distinguishes between real and fake news articles using machine learning techniques. Among the models tested, ensemble methods like Random Forest and Gradient Boosting demonstrated higher accuracy and robustness.

Future Enhancements

- Incorporating more advanced NLP models like BERT or LSTM.
 - Expanding the dataset for improved generalization.
 - Adding real-time scraping and classification capabilities.
 - Building a front-end for user interaction.
-